

Momentum Contrast for Unsupervised Visual Representation Learning

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Momentum Contrast for Unsupervised Visual Representation Learning

Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, Ross Girshick; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 9729-9738

Abstract

We present Momentum Contrast (MoCo) for unsupervised visual representation learning. From a perspective on contrastive learning as dictionary look-up, we build a dynamic dictionary with a queue and a moving-averaged encoder. This enables building a large and consistent dictionary on-the-fly that facilitates contrastive unsupervised learning. MoCo provides competitive results under the common linear protocol on ImageNet classification. More importantly, the representations learned by MoCo transfer well to downstream tasks. MoCo can outperform its supervised pre-training counterpart in 7 detection/segmentation tasks on PASCAL VOC, COCO, and other datasets, sometimes surpassing it by large margins. This suggests that the gap between unsupervised and supervised representation learning has been largely closed in many vision tasks.

Related Material

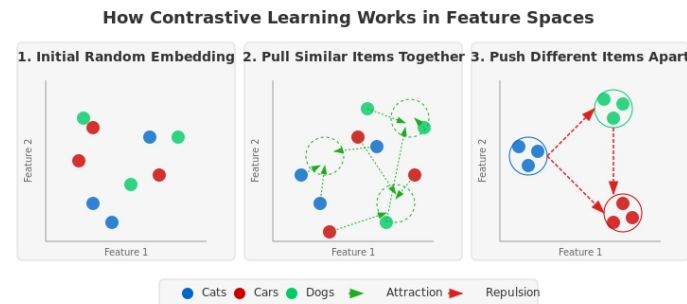
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- Introduction
- Related Work
- Method
- Experiments
- Discussion and Conclusion

- Unsupervised representation learning is highly successful in natural language processing.
- But supervised pre-training is still dominant in computer vision, where unsupervised methods generally lag behind.
- Why?
 - Language tasks have discrete signal spaces. -> easy to make dictionary.
 - Vision tasks have the raw signal is in a continuous, high-dimensional space.

- Recent Progress in Unsupervised Visual Representation Learning
 - Contrastive-loss-based methods show strong results.
 - Success depends on building an effective dynamic dictionary.
- Key Requirements for Contrastive Learning
 - Large Dictionary
 - Continuous visual space → many negatives needed.
 - Consistent Dictionary
 - Encoder for keys must change slowly to keep comparisons stable.



1. Queue-based Dictionary

- Enqueue current mini-batch keys
- Dequeue oldest keys
- Large, stable pool of negatives independent of batch size

2. Momentum Encoder

- Query encoder: Back prop.
- Key encoder: momentum update
- Ensures consistent key representations over time

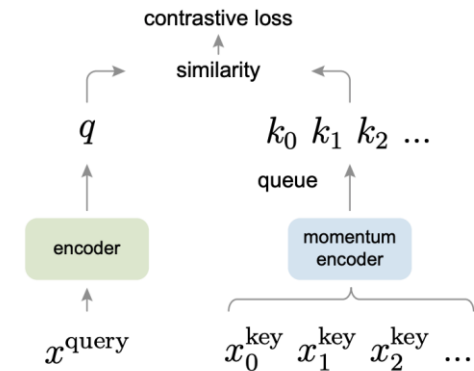
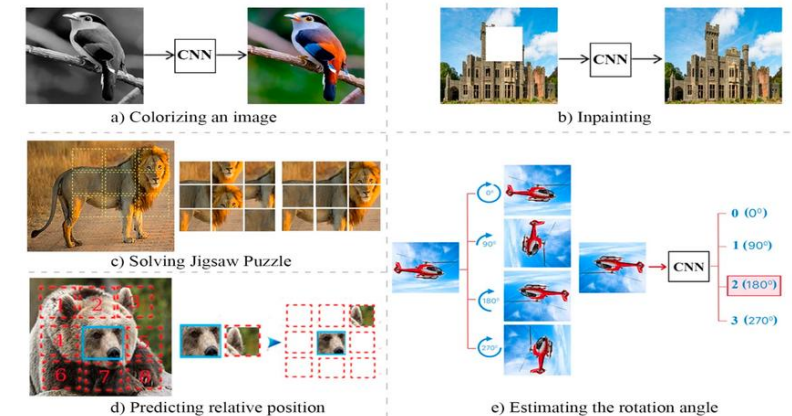


Figure 1. Momentum Contrast (MoCo) trains a visual representation encoder by matching an encoded query q to a dictionary of encoded keys using a contrastive loss. The dictionary keys $\{k_0, k_1, k_2, \dots\}$ are defined on-the-fly by a set of data samples. The dictionary is built as a queue, with the current mini-batch enqueued and the oldest mini-batch dequeued, decoupling it from the mini-batch size. The keys are encoded by a slowly progressing encoder, driven by a momentum update with the query encoder. This method enables a large and consistent dictionary for learning visual representations.

- Loss functions
 - Contrastive losses measure the “similarities” of sample pairs in a representation space.
 - Cosine similarity / noise-contrastive estimation(NCE) based
 - Adversarial losses measure the “difference” between probability distributions.
 - Data generation / KL Divergence
- Contrastive learning vs. pretext tasks
 - pretext tasks
 - Contrastive learning
 - Various pretext tasks can be based on some form of contrastive loss functions.



- InfoNCE loss

- Query ($q = f_q(x^q)$)
- Key ($k = f_k(x^k) / \{k_0, k_1, k_2, \dots\}$) / positive key (k^+)
- temperature hyper-parameter (τ)

$$L_q = -\log \frac{\exp\left(\frac{q \cdot k^+}{\tau}\right)}{\sum_{i=0}^K \exp\left(\frac{q \cdot k_i}{\tau}\right)}$$

- Intuitively, this loss is the log loss of a (K+1)-way softmax-based classifier that tries to classify q as k+.

$$p(C = \text{pos} | X, \mathbf{c}) = \frac{p(x_{\text{pos}} | \mathbf{c}) \prod_{i=1, \dots, N; i \neq \text{pos}} p(\mathbf{x}_i)}{\sum_{j=1}^N \left[p(\mathbf{x}_j | \mathbf{c}) \prod_{i=1, \dots, N; i \neq j} p(\mathbf{x}_i) \right]} = \frac{\frac{p(\mathbf{x}_{\text{pos}} | \mathbf{c})}{p(\mathbf{x}_{\text{pos}})}}{\sum_{j=1}^N \frac{p(\mathbf{x}_j | \mathbf{c})}{p(\mathbf{x}_j)}} = \frac{f(\mathbf{x}_{\text{pos}}, \mathbf{c})}{\sum_{j=1}^N f(\mathbf{x}_j, \mathbf{c})}$$

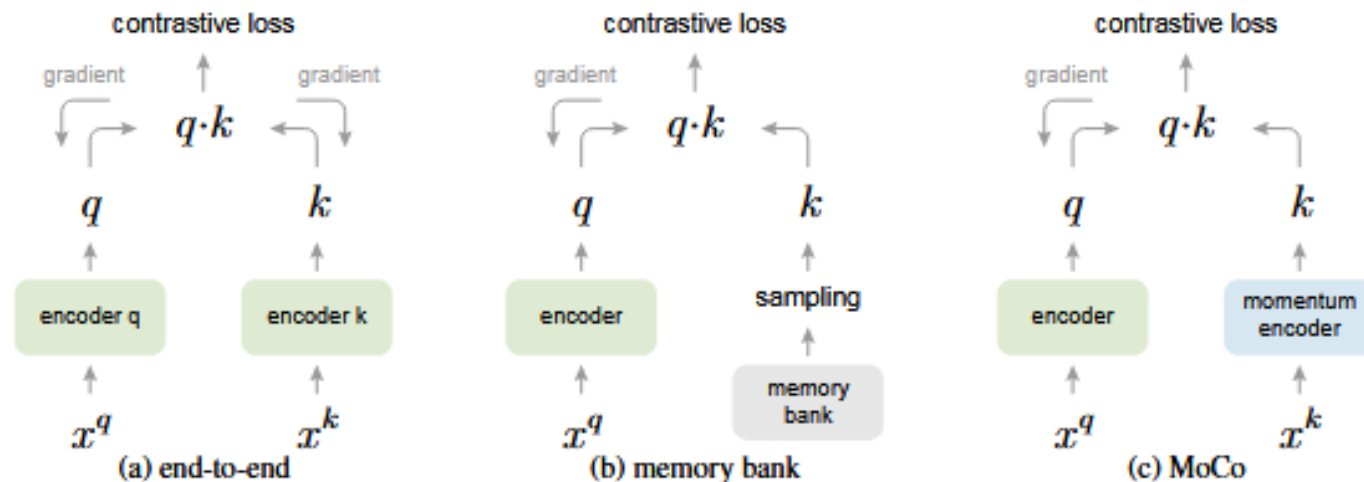
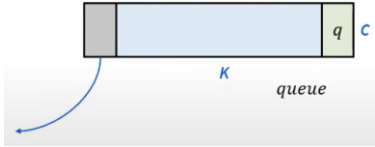


Figure 2. **Conceptual comparison of three contrastive loss mechanisms** (empirical comparisons are in Figure 3 and Table 3). Here we illustrate one pair of query and key. The three mechanisms differ in how the keys are maintained and how the key encoder is updated. (a): The encoders for computing the query and key representations are updated **end-to-end** by back-propagation (the two encoders can be different). (b): The key representations are sampled from a **memory bank** [61]. (c): **MoCo** encodes the new keys on-the-fly by a momentum-updated encoder, and maintains a queue (not illustrated in this figure) of keys.

- end-to-end
 - Dictionary: current mini-batch → keys consistently encoded
 - Limitation: dictionary size = mini-batch size
- memory bank
 - Dictionary: Stores all dataset representations → each key updated when processed
 - Limitation: keys encoded at different times → less consistent (limitation of random sampling)
- MoCo
 - Dictionary: Maintained as a queue.
 - Key consistency: Key encoder f_k is updated with momentum, ensuring that all keys in the queue remain similar to the most recent encoder.
 - Memory efficiency: Only the most recent mini-batches are stored instead of the entire dataset → scalable to large datasets.



Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

```
# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
    x_q = aug(x) # a randomly augmented version
    x_k = aug(x) # another randomly augmented version

    q = f_q.forward(x_q) # queries: NxK
    k = f_k.forward(x_k) # keys: NxK
    k = k.detach() # no gradient to keys

    # positive logits: Nx1
    l_pos = bmm(q.view(N,1,C), k.view(N,C,1))

    # negative logits: NxK
    l_neg = mm(q.view(N,C), queue.view(C,K))

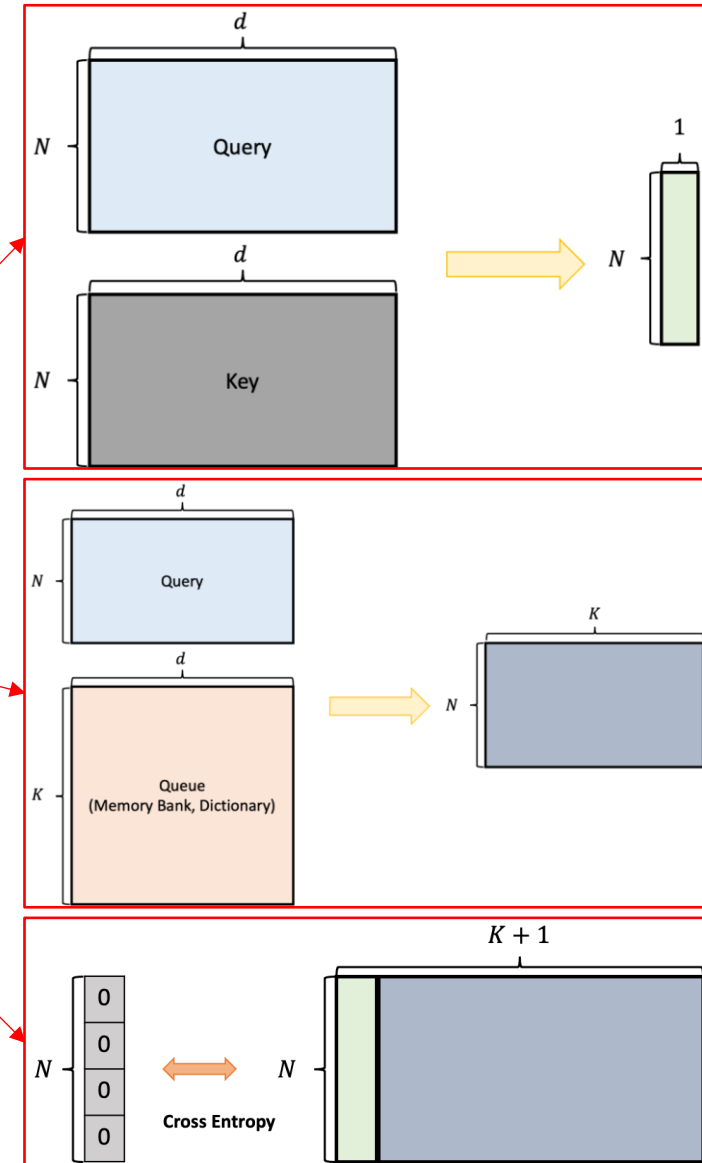
    # logits: Nx(1+K)
    logits = cat([l_pos, l_neg], dim=1)

    # contrastive loss, Eqn. (1)
    labels = zeros(N) # positives are the 0-th
    loss = CrossEntropyLoss(logits/t, labels)

    # SGD update: query network
    loss.backward()
    update(f_q.params)

    # momentum update: key network
    f_k.params = m*f_k.params+(1-m)*f_q.params
     $\theta_k \leftarrow m\theta_k + (1-m)\theta_q$ 
    # update dictionary
    enqueue(queue, k) # enqueue the current minibatch
    dequeue(queue) # dequeue the earliest minibatch
```

bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.



Dataset	Images	Characteristics	Mini-batch size	GPUs	Optimizer & Params	LR Schedule	Training Duration
ImageNet-1M (IN-1M)	~1.28M	Well-balanced, iconic objects	256	8	SGD W decay: 0.0001 Momentum: 0.9	Initial 0.03, $\times 0.1$ at 120 & 160 epochs	200 epochs ~53 hours (ResNet-50)
Instagram-1B (IG-1B)	~940M	Long-tailed, unbalanced, iconic scene-level images	1024	64	SGD W decay 0.0001 Momentum: 0.9	Initial 0.12, decayed $\times 0.9$ every 62.5k iters	1.25M iterations (~1.4 epochs) ~6 days (ResNet-50)

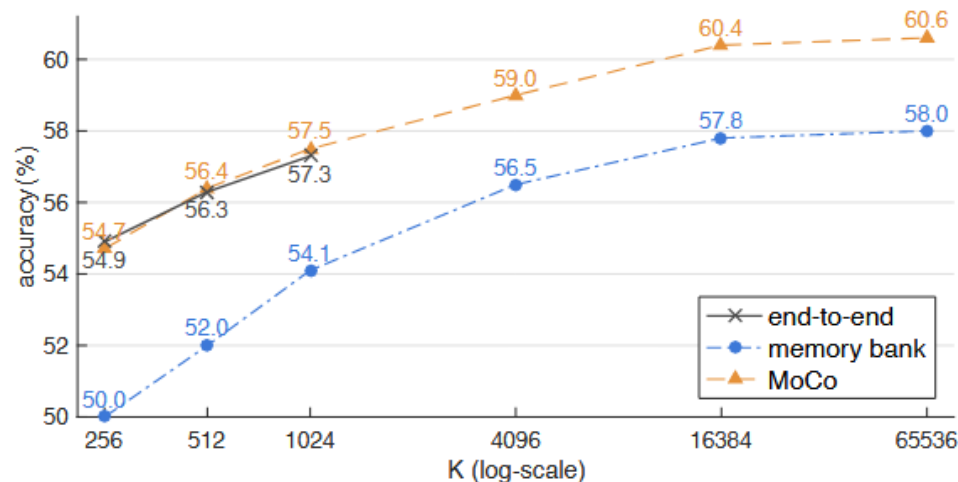
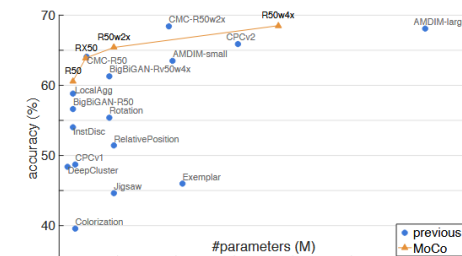


Figure 3. Comparison of three contrastive loss mechanisms under the ImageNet linear classification protocol. We adopt the same pretext task (Sec. 3.3) and only vary the contrastive loss mechanism (Figure 2). The number of negatives is K in memory bank and MoCo, and is $K-1$ in end-to-end (offset by one because the positive key is in the same mini-batch). The network is ResNet-50.

momentum m	0	0.9	0.99	0.999	0.9999
accuracy (%)	fail	55.2	57.8	59.0	58.9



method	architecture	#params (M)	accuracy (%)
Exemplar [17]	R50w3x	211	46.0 [38]
RelativePosition [13]	R50w2x	94	51.4 [38]
Jigsaw [45]	R50w2x	94	44.6 [38]
Rotation [19]	Rv50w4x	86	55.4 [38]
Colorization [64]	R101*	28	39.6 [14]
DeepCluster [3]	VGG [53]	15	48.4 [4]
BigBiGAN [16]	R50	24	56.6
	Rv50w4x	86	61.3
methods based on contrastive learning follow:			
InstDisc [61]	R50	24	54.0
LocalAgg [66]	R50	24	58.8
CPC v1 [46]	R101*	28	48.7
CPC v2 [35]	R170* _{wider}	303	65.9
CMC [56]	R50 _{L+ab}	47	64.1 [†]
	R50w2x _{L+ab}	188	68.4 [†]
AMDIM [2]	AMDIM _{small}	194	63.5 [†]
	AMDIM _{large}	626	68.1 [†]
MoCo	R50	24	60.6
	RX50	46	63.9
	R50w2x	94	65.4
	R50w4x	375	68.6

Table 1. Comparison under the linear classification protocol on ImageNet. The figure visualizes the table. All are reported as unsupervised pre-training on the ImageNet-1M training set, followed by supervised linear classification trained on frozen features, evaluated on the validation set. The parameter counts are those of the feature extractors. We compare with improved re-implementations if available (referenced after the numbers).

Notations: R101*/R170* is ResNet-101/170 with the last residual stage removed [14, 46, 35], and R170 is made wider [35]; Rv50 is a reversible net [23], RX50 is ResNeXt-50-32x8d [62].

[†]: Pre-training uses FastAutoAugment [40] that is supervised by ImageNet labels.

PASCAL VOC Object Detection

pre-train	AP ₅₀	AP	AP ₇₅
random init.	64.4	37.9	38.6
super. IN-1M	81.4	54.0	59.1
MoCo IN-1M	81.1 (-0.3)	54.6 (+0.6)	59.9 (+0.8)
MoCo IG-1B	81.6 (+0.2)	55.5 (+1.5)	61.2 (+2.1)

(a) Faster R-CNN, R50-dilated-C5

pre-train	AP ₅₀	AP	AP ₇₅
random init.	60.2	33.8	33.1
super. IN-1M	81.3	53.5	58.8
MoCo IN-1M	81.5 (+0.2)	55.9 (+2.4)	62.6 (+3.8)
MoCo IG-1B	82.2 (+0.9)	57.2 (+3.7)	63.7 (+4.9)

(b) Faster R-CNN, R50-C4

Table 2. Object detection fine-tuned on PASCAL VOC trainval07+12. Evaluation is on test2007: AP₅₀ (default VOC metric), AP (COCO-style), and AP₇₅, averaged over 5 trials. All are fine-tuned for 24k iterations (~23 epochs). In the brackets are the gaps to the ImageNet supervised pre-training counterpart. In green are the gaps of at least +0.5 point.

pre-train	R50-dilated-C5			R50-C4		
	AP ₅₀	AP	AP ₇₅	AP ₅₀	AP	AP ₇₅
end-to-end	79.2	52.0	56.6	80.4	54.6	60.3
memory bank	79.8	52.9	57.9	80.6	54.9	60.6
MoCo	81.1	54.6	59.9	81.5	55.9	62.6

Table 3. Comparison of three contrastive loss mechanisms on PASCAL VOC object detection, fine-tuned on trainval07+12 and evaluated on test2007 (averages over 5 trials). All models are implemented by us (Figure 3), pre-trained on IN-1M, and fine-tuned using the same settings as in Table 2.

pre-train	AP ₅₀				AP MoCo	AP ₇₅	
	RelPos, by [14]	Multi-task [14]	Jigsaw, by [26]	LocalAgg [66]		Multi-task [14]	MoCo
super. IN-1M	74.2	74.2	70.5	74.6	74.4	44.3	42.7
unsup. IN-1M	66.8 (-7.4)	70.5 (-3.7)	61.4 (-9.1)	69.1 (-5.5)	74.9 (+0.5)	43.9 (-0.4)	50.1 (+7.4)
unsup. IN-14M	-	-	69.2 (-1.3)	-	75.2 (+0.8)	-	50.2 (+7.5)
unsup. YFCC-100M	-	-	66.6 (-3.9)	-	74.7 (+0.3)	-	49.0 (+6.3)
unsup. IG-1B	-	-	-	-	75.6 (+1.2)	-	51.7 (+9.0)

Table 4. Comparison with previous methods on object detection fine-tuned on PASCAL VOC trainval2007. Evaluation is on test2007. The ImageNet supervised counterparts are from the respective papers, and are reported as having the same structure as the respective unsupervised pre-training counterparts. All entries are based on the C4 backbone. The models in [14] are R101 v2 [34], and others are R50. The RelPos (relative position) [13] result is the best single-task case in the Multi-task paper [14]. The Jigsaw [45] result is from the ResNet-based implementation in [26]. Our results are with 9k-iteration fine-tuning, averaged over 5 trials. In the brackets are the gaps to the ImageNet supervised pre-training counterpart. In green are the gaps of at least +0.5 point.

COCO segmentation

COCO keypoint detection				
pre-train	AP ^{kp}	AP ^{kp} ₅₀	AP ^{kp} ₇₅	
random init.	65.9	86.5	71.7	
supcr. IN-1M	65.8	86.9	71.9	
MoCo IN-1M	66.8 (+1.0)	87.4 (+0.5)	72.5 (+0.6)	
MoCo IG-1B	66.9 (+1.1)	87.8 (+0.9)	73.0 (+1.1)	

COCO dense pose estimation			
pre-train	AP ^{dp}	AP ^{dp} ₅₀	AP ^{dp} ₇₅
random init.	39.4	78.5	35.1
supcr. IN-1M	48.3	85.6	50.6
MoCo IN-1M	50.1 (+1.8)	86.8 (+1.2)	53.9 (+3.3)
MoCo IG-1B	50.6 (+2.3)	87.0 (+1.4)	54.3 (+3.7)

LVIS v0.5 instance segmentation			
pre-train	AP ^m	AP ^m ₅₀	AP ^m ₇₅
random init.	22.5	34.8	23.8
supcr. IN-1M†	24.4	37.8	25.8
MoCo IN-1M	24.1 (−0.3)	37.4 (−0.4)	25.5 (−0.3)
MoCo IG-1B	24.9 (+0.5)	38.2 (+0.4)	26.4 (+0.6)

Cityscapes instance seg.		Semantic seg. (mIoU)	
pre-train	AP ^m	AP ^m ₅₀	VOC
random init.	25.4	51.1	65.3
supcr. IN-1M	32.9	59.6	74.6
MoCo IN-1M	32.3 (−0.6)	59.3 (−0.3)	75.3 (+0.7)
MoCo IG-1B	32.9 (0.0)	60.3 (+0.7)	75.5 (+0.9)

Table 6. MoCo vs. ImageNet supervised pre-training, fine-tuned on various tasks. For each task, the same architecture and schedule are used for all entries (see appendix). In the brackets are the gaps to the ImageNet supervised pre-training counterpart. In green are the gaps of at least +0.5 point.

†: this entry is with BN frozen, which improves results; see main text.

pre-train	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅
random init.	31.0	49.5	33.2	28.5	46.8	30.4	36.7	56.7	40.0	33.7	53.8	35.9
supcr. IN-1M	38.9	59.6	42.7	35.4	56.5	38.1	40.6	61.3	44.4	36.8	58.1	39.5
MoCo IN-1M	38.5 (−0.4)	58.9 (−0.7)	42.0 (−0.7)	35.1 (−0.3)	55.9 (−0.6)	37.7 (−0.4)	40.8 (+0.2)	61.6 (+0.3)	44.7 (+0.3)	36.9 (+0.1)	58.4 (+0.3)	39.7 (+0.2)
MoCo IG-1B	38.9 (0.0)	59.4 (−0.2)	42.3 (−0.4)	35.4 (0.0)	56.5 (0.0)	37.9 (−0.2)	41.1 (+0.5)	61.8 (+0.5)	45.1 (+0.7)	37.4 (+0.6)	59.1 (+1.0)	40.2 (+0.7)

(a) Mask R-CNN, R50-FPN, 1× schedule

pre-train	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅
random init.	26.4	44.0	27.8	29.3	46.9	30.8	35.6	54.6	38.2	31.4	51.5	33.5
supcr. IN-1M	38.2	58.2	41.2	33.3	54.7	35.2	40.0	59.9	43.1	34.7	56.5	36.9
MoCo IN-1M	38.5 (+0.3)	58.3 (+0.1)	41.6 (+0.4)	33.6 (+0.3)	54.8 (+0.1)	35.6 (+0.4)	40.7 (+0.7)	60.5 (+0.6)	44.1 (+1.0)	35.4 (+0.7)	57.3 (+0.8)	37.6 (+0.7)
MoCo IG-1B	39.1 (+0.9)	58.7 (+0.5)	42.2 (+1.0)	34.1 (+0.8)	55.4 (+0.7)	36.4 (+1.2)	41.1 (+1.1)	60.7 (+0.8)	44.8 (+1.7)	35.6 (+0.9)	57.4 (+0.9)	38.1 (+1.2)

(b) Mask R-CNN, R50-FPN, 2× schedule

pre-train	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅
random init.	26.4	44.0	27.8	29.3	46.9	30.8	35.6	54.6	38.2	31.4	51.5	33.5
supcr. IN-1M	38.2	58.2	41.2	33.3	54.7	35.2	40.0	59.9	43.1	34.7	56.5	36.9
MoCo IN-1M	38.5 (+0.3)	58.3 (+0.1)	41.6 (+0.4)	33.6 (+0.3)	54.8 (+0.1)	35.6 (+0.4)	40.7 (+0.7)	60.5 (+0.6)	44.1 (+1.0)	35.4 (+0.7)	57.3 (+0.8)	37.6 (+0.7)
MoCo IG-1B	39.1 (+0.9)	58.7 (+0.5)	42.2 (+1.0)	34.1 (+0.8)	55.4 (+0.7)	36.4 (+1.2)	41.1 (+1.1)	60.7 (+0.8)	44.8 (+1.7)	35.6 (+0.9)	57.4 (+0.9)	38.1 (+1.2)

(c) Mask R-CNN, R50-C4, 1× schedule

pre-train	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^m	AP ^m ₅₀	AP ^m ₇₅
random init.	26.4	44.0	27.8	29.3	46.9	30.8	35.6	54.6	38.2	31.4	51.5	33.5
supcr. IN-1M	38.2	58.2	41.2	33.3	54.7	35.2	40.0	59.9	43.1	34.7	56.5	36.9
MoCo IN-1M	38.5 (+0.3)	58.3 (+0.1)	41.6 (+0.4)	33.6 (+0.3)	54.8 (+0.1)	35.6 (+0.4)	40.7 (+0.7)	60.5 (+0.6)	44.1 (+1.0)	35.4 (+0.7)	57.3 (+0.8)	37.6 (+0.7)
MoCo IG-1B	39.1 (+0.9)	58.7 (+0.5)	42.2 (+1.0)	34.1 (+0.8)	55.4 (+0.7)	36.4 (+1.2)	41.1 (+1.1)	60.7 (+0.8)	44.8 (+1.7)	35.6 (+0.9)	57.4 (+0.9)	38.1 (+1.2)

(d) Mask R-CNN, R50-C4, 2× schedule

Table 5. Object detection and instance segmentation fine-tuned on COCO: bounding-box AP (AP^{bb}) and mask AP (AP^m) evaluated on val2017. In the brackets are the gaps to the ImageNet supervised pre-training counterpart. In green are the gaps of at least +0.5 point.

- Positive results of unsupervised learning in a variety of computer vision tasks and datasets.
- MoCo's improvement from IN-1M to IG-1B is consistently noticeable but relatively small
 - suggesting that the larger-scale data may not be fully exploited.
 - We hope an advanced pretext task will improve this.
- Beyond the simple instance discrimination task.
 - it is possible to adopt MoCo for pretext tasks like masked auto-encoding.